

Six Activations in Twelve Users

AP Statistics · Unit 3 · Topic 3.3 · TEACHER KEY

CED TETHER

- **VAR-1.H** Identify questions suggested by variation in the shapes of distributions of samples taken from the same population. [Skill 1.A]
- **VAR-1.H.1** Variation in shapes of data distributions may be random or not.

Essential question: When should a simulated sampling distribution be trusted more than a normal approximation?

DOK-3 skill: 4.B · **FRQ pattern:** normal-approximation-vs-simulation-for-a-skewed-tail

DOK rationale: Part (c) coordinates an expected-count check, the simulated distribution's shape, and two tail estimates to choose a defensible method, bound the evidence, and explain the consequence of the rejected analysis.

Self-paced bonus sheet. Students take it when they choose and turn it in when they finish. Everything they need is printed on the sheet. Score part (c) only, E / P / I, by hand — never AI-graded or auto-scored. (a) and (b) are the ladder up; use them to see where a thin (c) came from.

Questions to ask

- What does one bar in this simulated distribution count?
- Which expected count warns you that a symmetric curve may miss the tail?
- Why do simulated samples with more than six activations still count?
- Can the two methods disagree in size yet lead to the same 5 percent decision?

[WARN] LOOK FOR

- Treating every sampling distribution as normal regardless of expected counts.
- Counting exactly six activations instead of six or more.
- Saying a small chance probability proves why the activation rate differs.

SCORING PART (c) — E / P / I

Expected elements

- **judgment-1** (required) — Chooses the simulation using failed expected counts 2.4 and 9.6 or the right-skewed shape.
- **judgment-2** (required) — Uses $4/200=0.020$ below 0.05 as evidence against the model, without claiming proof.
- **judgment-3** (required) — Explains that 0.005 makes the result seem four times rarer and overstates the evidence.

E	Meets all three checks with correct contextual reasoning; equivalent wording is acceptable.
P	Shows at least one substantive correct check, but has an omission or error that prevents E.
I	Shows none of the three checks with substantive correct reasoning; a choice alone is insufficient.

Common mistakes

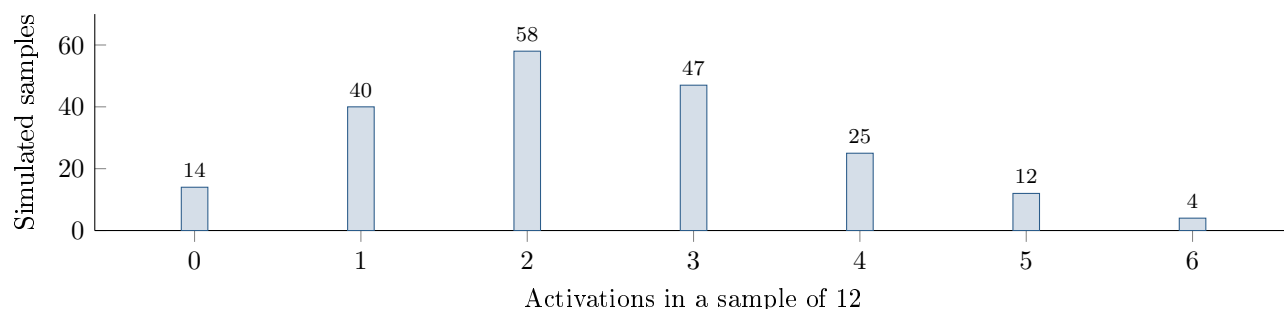
- Using $6/12 = 0.50$ as the chance probability
- Counting only the bar at 6 without recognizing that all larger results would also be at least as extreme
- Checking observed successes and failures instead of expected counts under p_0 for this model
- Calling the two percent estimate proof that the developer intentionally misstated the rate

[STAR] THE PROBLEM — annotated

Suppose a developer’s model says $p_0 = 0.20$ of new app users activate voice commands. An SRS of 12 users includes 6 activations, so $\hat{p} = 0.50$. The chart shows 200 simulated samples of 12 independent users under this model.

Mara: “A normal model for $\hat{p}$ gives $P(\hat{p} \geq 0.50) \approx 0.005$, so that is the chance probability I would report.”

Jules: “The simulated distribution is not very normal. I would estimate the chance probability directly from its tail.”



FIRST TAKE — one sentence before you work the parts

Whose method would you trust more, Mara’s smooth curve or Jules’s repeated trials? Give one reason from the picture.

Key: Not graded. Preserve the normal-model versus matched-simulation tension.

[BOOK] CHECK THE SHAPE BEFORE USING NORMAL (keep on the page)

Under a proposed proportion p_0 , check expected counts $np_0 \geq 10$ and $n(1 - p_0) \geq 10$ before using a normal approximation. A matched simulation estimates the chance probability by the number of simulated results at least as large as the observed result, divided by the total simulations. For this task, a chance probability below 5% counts as convincing evidence against the proposed model. Evidence against a model does not prove a particular cause.

DOK 1 (a) State the most common number of activations in the simulated samples and describe the distribution as roughly symmetric, skewed left, or skewed right.

Key: Two activations is most common (58 simulations). The distribution is skewed right.

DOK 2 (b) Use the chart to estimate $P(\hat{p} \geq 0.50)$ under $p_0 = 0.20$. Then calculate np_0 and $n(1 - p_0)$ and assess the expected-count condition for a normal approximation.

Key: The simulated tail is $4/200 = 0.020$. Expected counts $12(0.20) = 2.4$ and $12(0.80) = 9.6$ both fall below 10, so the normal condition fails.

DOK 3 (c) In 3–4 sentences, choose a method using the simulated tail and either the shape or expected counts. Apply the 5% evidence rule. Explain how reporting 0.005 instead of 0.020 would change a reader’s impression.

Key: Use Jules’s simulation: the failed expected counts and right skew weaken the normal approximation. The tail $0.020 < 0.05$ gives evidence against the model, without proving a cause. Reporting 0.005 makes the result seem four times rarer and overstates the evidence.